CMS Health Technology Ecosystem RFI

We at Curai Health would like to thank CMS for inviting public comment on the Health Technology Ecosystem RFI. Curai Health is an AI-enabled and driven virtual primary care service with a mission to bring the world's best healthcare to every human being on the planet. As a care-providing organization with a heavy focus on technology, we have provided responses to the questions targeted at patient needs, provider groups, and technology vendors. Our responses are highlighted in green below the questions they pertain to. We look forward to working with CMS to continue to drive the health technology ecosystem forward.

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1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

Digital health products that deliver care, help make it more accessible, or consolidate information for personal use would help understand and manage health needs for myself and my loved ones.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

Digital health products can enable the ability to have consistent and convenient access to various care services (ranging from virtual urgent care to continuous primary care to care navigation and coordination to chronic condition management) in one's pocket. Digital health services could also allow for management of care and monitoring of health for loved ones.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

The most valuable tasks that a personal assistant could do to support one's health needs is in taking on care coordination burden (appointment scheduling, paperwork) and in care navigation (finding an in-network provider or specialist, etc).

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

No - in fact this fragmentation is commonly a significant barrier to helping provide oversight or care for loved ones. And even on an individual basis, records are split between various

systems, especially when utilizing different healthcare providers for different services (e.g. in-person urgent care visit with a hospital system, telemedicine visit with a healthtech company, vaccine appointment with the state).

a. If so, what are some examples of benefits it has provided?

In (rare) situations where data is accurately transmitted and synchronized between provider groups, I as a patient can track my health in a unified spot. I have almost never seen a provider use this information proactively.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

As a patient, this would be a hyper-valuable asset to bring to a doctor's appointment (to show a clinician a full and complete picture of one's health). This would also be very valuable when changing care providers or moving, and directly replace a box full of paperwork / faxes from the source to the target provider.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

Even with current technologies and interoperability pushes, imaging data, claims data, and encounter notes are almost never available for consumption (in contrast to prescriptions, lab results, and electronic after-visit summaries). This is a limitation in functionality with many EHR systems (though FHIR models and APIs for these data elements exist). In general, the biggest obstacle for obtaining and managing this information is provider systems being willing to expose this information in standards-compliant ways. As a concrete example of this, Apple Health can ingest large swathes of FHIR data, but only if health systems choose to support exporting to Apple Health via SMART on FHIR API connectors.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Curai Health serves as a unified AI-enabled virtual primary care app that helps fill care gaps and is especially convenient for patients in rural areas or those with barriers to accessing care. Curai also provides chronic condition management and care coordination / navigation services.

PC-4. What features are missing from apps you use or that you are aware of today?

Most healthcare applications are siloed, in that they have reasonable context into anything that happens on their platforms, but generally struggle to connect patients with healthcare-related events that happen off-platform. Increased emphasis on interoperability, as well as reducing the burden of pulling in these resources, would greatly improve these apps.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Many digital health and virtual care applications do not serve Medicare or Medicaid populations. A big component of this absence is the fact that getting ready to serve these populations is onerous on various fronts (certification, approval, compliance, reimbursement), which is something that CMS could choose to speed up.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

Given CMS's Medicare and Medicaid populations, it is uniquely positioned to embrace digital health and virtual care applications. Many of the groups served by Medicare and Medicaid have to deal with access and convenience issues, and are (typically) plagued by chronic conditions that need constant touch points and feedback loops to manage (which are very expensive to handle in a brick-and-mortar environment). Digital health and virtual care (and technology more broadly) can help these patients by meeting them where they are and leveraging constant touch points to make them healthier.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

Outreach campaigns, promotions, and incentives would be beneficial in encouraging patient and caregiver interest in these products. It is also worth noting that these products, when used in the right circumstances, increase care access and provide convenience for groups that can materially benefit from increased access and convenience.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

In our opinion, putting digital health products through approval processes will serve to stifle innovation, adoption, and reduce positive health outcomes. However, reviewing applications for efficacy / quality / impact on health outcomes retroactively is something that should be done to make sure that endorsed applications are appropriately serving the needs of patients. Communication with patients, simpler data access, and reimbursement adjustments, as suggested by the question, would all serve to increase patient and caregiver adoption of digital health products.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

A standards-compliant information exchange for CMS data (a la the FHIR APIs seen here: https://developer.cms.gov/), with the ability for non-CMS provider groups to access CMS data for patients under their care, would be beneficial to enabling timely access to this data. An

alternative approach would be for CMS to mandate that participating provider groups push data via existing channels into health information exchanges, thus making said data accessible to any group subscribed to the exchange.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Ease of use and onboarding, accessibility, simplicity of user interface and experience, and support for multiple modalities (voice, text, video) are key requirements to making digital health products accessible and easy to use for Medicare beneficiaries and caregivers.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

Measuring health-related outcomes via value-based metrics feels like the best means of doing so.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

The most readily-available information for patients, caregivers, and providers alike are lab data (and corresponding observations), immunizations, vitals, and medication records.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

The most difficult data elements to access are all listed in the question - claims data, encounter notes, operative reports, appointment schedules, and prices are all not generally available.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

Encounter notes would be of immense volume, especially with the abilities of large language models (LLMs) and other AI / machine learning tools to interpret, summarize, and utilize these notes.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

The biggest barriers and challenges are standardization and adoption. There are many sources of clinical data that are simply not available for researchers, vendors, or provider groups to access and utilize in standard ways; and if some groups adopt the standards, others don't (or subscribe to alternate standards).

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

As mentioned above, clinical data and encounter notes would be incredibly valuable to enable more useful digital health products.

a. What difficulties are there in accessing or utilizing these data sources today?

Many provider systems do not publish this information in any sort of programmatically accessible or consistent fashion - those that do often fall back to using binary data (PDFs) to push this information. This makes utilization more complicated than it strictly needs to be.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

Increasing the kinds of data returned by the Blue Button 2.0 API (as mentioned above), and improving the ergonomics of the authorization flow (which involves complexity on both the developer and the patient) would improve the experience.

c. Is there non-CMS data that should be included in the API?

Yes - in particular, finding ways to include clinical information (Encounters, DocumentReferences, Observations, etc.) and potentially including information shared via health information exchanges would be crucial additions to the API.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

At present, it is unclear if this is making a difference yet (though the potential is there).

a. Please provide specific examples.

As TEFCA remains optional, there is not a guarantee that any specific healthcare entity is a participant (or connected to someone who is). Thus, from the perspective of a patient attempting to access their health information in the real world, there isn't really a clear pathway to utilize TEFCA.

b. What changes would you suggest?

Continuing to push for broader participation in TEFCA and aligning TEFCA with other standards to make participation more straightforward would both help drive uptake and eventual patient impact.

c. What use cases could have a significant impact if implemented through TEFCA?

As mentioned below in PC-12a, scheduling, price transparency, and orders would all have significant impact.

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

FHIR standards and health information exchanges are both working well to advance access. The FHIR specification has allowed for health records to be shared efficiently (and even be integrated into systems like Apple Health). Health information exchanges are allowing information from provider groups to transmit more seamlessly (though not perfectly) between each other.

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

The standards that could improve data access and integration are already seeing good use (see: HIEs, FHIR / USCDI, etc).

f. Are there redundant standards, protocols, or channels that should be consolidated?

Yes - as mentioned in PR-6d and TD-6b, there is heavy overlap between FHIR + associated standards like Smart-on-FHIR, FHIR profiling systems like USCDI / QICore, and TEFCA. In addition, there is also redundancy between functionality in FHIR and legacy functionality in HL7 v2. Reducing the number of competing frameworks here and consolidating into fewer paths will be beneficial.

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

Partially - direct Smart-on-FHIR access to supported health systems and plans allows patients to easily see and control their own health information, but support for this is not comprehensive. TEFCA will help push on this.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

Most HIEs support patients requesting their own data, or provide healthcare provider groups the ability to request patient consent to have a provider system access said patient's HIE data.

a. How valuable, available, and accurate do you find the data they share to be?

In general, very valuable and available, with accuracy depending on the type of data being shared. Data with normative status tends to be more accurate than less normative data.

b. What changes would you suggest?

Continuing to push for reciprocity from groups pushing / pulling from HIEs, and further normalization of data standards across HIEs, would be helpful changes in further improving the quality and availability of healthcare data.

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

The highest performing HIE models we have seen transact in FHIR, apply standardization rules at the point of pushing into the network, and work to simplify downstream work for provider groups and consumers.

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

There is clearly overlap between these various groups - in our opinion, having HIEs take the next step and become QHINs would reduce confusion and complexity in this crucial area.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

a. Examples may include the following:

(1) Binding cost estimates for pre-defined periods.

(2) Viewing provider schedule availability.

(3) Using third-party apps for appointment management.

(4) Accessing patient-facing quality metrics.

(5) Finding the right provider for specific healthcare needs.

In our opinion, the most valuable use cases are also the ones that are most impossible today - meaning price transparency, scheduling, provider directory lookups, and orders (lab and RX). Currently, scheduling and orders in particular require bespoke integrations and connections, which greatly complicates care navigation.

b. What use cases are possible today?

At present, the only use cases that are truly possible are pulling and pushing information from information exchanges.

c. What should be possible in the near future?

There are vendors that have come up with elegant solutions for ordering, but these are vendor specific. There are international examples of ordering workflows being standardized (see: UK and Norway) - so this should be possible if backed by the appropriate resources.

d. What would be very valuable but may be very hard to achieve?

Scheduling would make care navigation a breeze, but at present is incredibly unique on a per-provider-system basis. Solving provider availability / scheduling / appointment booking in an interoperable, consistent way would make a huge difference in terms of impactful operational health data use cases.

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

Better publicizing the portal and making it even easier to submit blocking complaints would encourage broader usage and utilization. Increasing the public reporting of complaints should positively impact data exchange.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

The biggest challenges with digital identity credentials are usability, especially with less-technically-savvy populations, as well as fragmentation - each of the digital identity solutions mentioned above sees practical real-world use, and one needs to remember passwords for all of them.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

Increased usage of digital identity credentials by patients and caregivers would theoretically allow for patient identity verification, and could serve as a mechanism to uniquely identify an individual without relying on patient-reported metadata.

c. What are the potential downsides?

As mentioned above, usability and fragmentation (as well as a heavy reliance on critical third-party vendors) are a potentially large downside. Many individuals are uncomfortable with the concept of a unified digital identity profile being a prerequisite for access to care; in addition, many of the populations served by CMS / Medicare / Medicaid may not be the most comfortable with these digital ID solutions (which require credentials as well as multi-factor auth and / or passkeys).

d. How would encouraging the use of CSPs improve access to health information?

Given the benefits and downsides above, encouraging the use of CSPs, coupled with digital identity credentials being accepted as a means to access health information, would certainly improve access, with a large caveat that the downsides listed above still apply in force.

e. What role should CMS/payers, providers, and app developers have in driving adoption?

CMS, payers, and app developers could strongly suggest (or even mandate) that patients sign up for digital identity credentials before receiving care. Again, the downsides of a mandate should be weighed against the benefits.

f. How can CMS encourage patients to get digital identity credentials?

CMS can incentivize this via outreach / marketing, or in a more rigid fashion mandate this as a precursor to digital (not physical) care.

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

CMS and partners can incentivize provider groups to leverage digital health products as a part of the end-to-end care journeys of patients. This would be impactful in general, but especially impactful in rural areas where access to care is a significant barrier to positive health outcomes.

a. What are the current obstacles?

Digital health companies generally stay away from working with CMS and partners because of onerous certification requirements, high barriers to entry, and unclear pathways for reimbursement (particularly if a digital health company provides virtual care). Finding ways to

expedite approval / certification processes and clarifying reimbursement pathways would go a long way towards addressing the current obstacles.

b. What information should providers share with patients when using digital products in the provision of their care?

Providers should share (and know) the reasons for recommending a particular digital product and any privacy / security / data sharing considerations that are involved in the provisioning and use of the digital product(s). In addition, particularly when working with populations that may not have the most advanced technical literacy, it would also be beneficial for providers to walk patients through how to use the digital products they recommend.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

Providers must ensure that the digital product serves a clear care need, and that they are able to articulate what the product does in full.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

The biggest obstacles to developing and deploying innovative applications for physician workflows are closed EHR systems, which do not expose externally the APIs and programmatic interfaces that they utilize internally for all functions. Mandating that provider systems (from intake through clinical notes through to billing) have open APIs would go a long way towards mitigating these obstacles and allowing for useful and innovative applications to be developed and deployed.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

As a technology company and a provider network, it is crucial that all data in a system be accessible for exchange, even if formats are not fully standardized (though some data elements are more important than others, see also PC-8a and PC-8b).

a. Current challenges in accessing different data formats.

Historically, analyzing unstructured data has been a big barrier to utilizing different data formats. With modern advances in OCR, image processing / recognition, and natural language parsing and understanding, many of these issues are either mostly or fully mitigated, which then pushes the barrier to *pushing* these different formats into exchanges.

b. Impact on patient care quality.

It is well understood that a full, holistic understanding of a patient's healthcare journey can lead to more personalized care and better health outcomes as a result. Having access to *all* of a patient's records (even those that are scanned / faxed / unstructured) helps build this holistic understanding, and thus should be highly prioritized.

c. Technical barriers to full data accessibility.

Given the advances detailed above, the biggest technical barrier falls to EHR systems to find avenues to push this unstructured data into exchanges (and for exchange protocols to support these types of data elements). Given that FHIR (and similar) protocols support file uploads as a fallback, this is mostly solved on the exchange side.

d. Cost or privacy implications of making all data formats interoperable.

There is a cost to EHR systems to push this data into exchanges, and a slightly larger digital surface area (albeit unstructured) for a patient, which is a privacy implication; but the upside is that a patient's care journey is significantly more continuous across provider groups.

e. Priority level compared to other interoperability needs.

This is certainly an important interoperability area, but in our eyes secondary to enforcing fully-featured open (EHR) APIs.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

The types of administrative workflows and enhancements detailed here are also dependent on open, standardized EHR APIs - in particular, intake forms, provider information, and appointment scheduling are all dependent on provider EHR systems exposing this functionality. For example, there is a formal appointment-booking workflow detailed in the FHIR spec, but the implementations of this workflow across the most popular EHR providers do not conform to the spec and require heavy coordination with individual provider systems (and thus are not-at-all generic).

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

Curai Health runs a FHIR-native EHR, where all clinical data is stored in the FHIR format (US Core and QI Core minimum compliance).

a. Patient Access API.

We support all patient access APIs, with both individual and bulk transactions.

b. Standardized API for Patient and Population Services.

Yes - we support the standardized API for patient and population services by virtue of deploying a fully standards-compliant FHIR server and profiling our resources to US Core at a minimum.

c. Provider Directory API.

We collect and store all of the data required to power the Provider Directory API in FHIR, but do not currently expose this API externally.

d. Provider Access API.

As we are not a payer, we do not expose this API, but have the ability to consume it when impacted health plans and payer organizations begin supporting it.

e. Payer-to-Payer API.

As we are not a payer, we do not expose this API, but have the ability to consume it when impacted health plans and payer organizations begin supporting it.

f. Prior Authorization API.

We connect to and utilize these APIs through vendor partners.

g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries?

Yes, we support all bulk FHIR operations, including Group ID-based access filtering.

h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?

As our FHIR data is meant for internal use, we support EHR-launched app access (with internally controlled deployments). We intend to support standalone app access in the future.

i. CDS Hooks (for clinical decision support integrations).

We utilize CDS hooks to power our internal clinical decision support modules.

PR-6. Is TEFCA currently helping to advance provider access to health information?

This remains to be seen.

a. Please provide specific examples.

As currently written, TEFCA is an opt-in for institutions choosing to become qualified health information networks - this list currently includes a handful of major players (CommonWell, Epic, Health Gorilla, SureScripts) but also notably excludes a number of other major players. In addition, TEFCA has FHIR-based APIs on the roadmap, but this is still in draft.

b. What changes would you suggest?

TEFCA should reduce the certification burden of becoming a QHIN and commit fully to FHIR, which has broad industry-wide support. In addition, there are a number of workflows that are still not standardized (appointment booking, prescription and lab ordering referrals) - for interoperability to extend beyond just data transfer and to clinical operations themselves, additions to the standard are needed.

c. What other options are available outside of TEFCA?

Direct point-to-point integrations, using vendors like Health Gorilla, or engaging with networks like CommonWell and CareQuality all remain options.

d. Are there redundant standards, protocols or channels or both that could be consolidated?

Yes - in particular, FHIR + Smart on FHIR, TEFCA, USCDI, and similar are all *just* adjacent to each other. In addition, HL7 v2 protocols remain in place and heavily used, which serves as a barrier to modernization with FHIR.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

Providers are always looking for a complete picture of a patient's health journey, and as such are incentivized to utilize information exchanges to pull health information that may have been created outside of their own system. Utilizing these existing paved pathways, as well as emphasizing reciprocity, would continue to improve interoperability and mitigate the burden of sharing.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

CMS (or partners) could publish (or host) open source tools and frameworks for computing derived clinical quality data from source (likely FHIR). This would standardize metric computations and also incentivize provider groups to publish data in a format that is readily usable by these tools and frameworks.

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

Bulk FHIR data exports from EHRs to CMS would have the benefit of offloading clinical quality data submission calculations to CMS, with the downside of needing to potentially implement more FHIR APIs, ensure standards-compliant FHIR formatting, and dealing with auth key exchanges. In our view, this would indeed reduce the burden on provider groups, despite the tradeoffs.

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

As mentioned above, if quality reporting was able to be done automatically against a provider group that publishes the information required for these computations in an interoperable manner (e.g. via FHIR through an information network), then the provider's responsibility is simply to interoperate, which clearly dual-purposes the investment.

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

We believe that CMS and partners should continue to push for real-time (or near-real-time) access to data in general, not just on the quality data front. This would allow for more-informed care delivery workflows for provider groups and improve the quality of care dispensed overall.

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

CMS could incentivize providers who accept approved digital identity credentials, or conversely penalize those who do not.

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

If the single set of trusted credentials involves multiple different vendors, provider groups would benefit greatly from a singular standard that unifies the vendors except for minor configuration differences (for example, SSO via OAuth 2).

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

As evidenced by the public comments already submitted for this RFI, the biggest barrier will be patient adoption, distrust of these identity providers, and a lot of misunderstandings as to what digital identity credentials really mean. Education and outreach will be very important to addressing patient privacy concerns.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers?

The clear benefits are in single sign-on capabilities across multiple systems for providers, with a downside being the security risks with one master credential that, if compromised, would grant unfettered access to a bad actor (this is largely mitigated by MFA standards, though not fully so).

b. How would requiring their use improve access to health information?

From a provider's perspective, the assumption would be that this digital identity credential could be used across multiple platforms and systems to grant access to necessary patient and population information.

c. What are the potential downsides?

This places a large implementation burden on each platform and system to accept these credentials and tie them to the appropriate levels of permissions and access.

d. What impact would mandatory credentials have on a nationwide provider directory?

In our opinion, simply mandating digital identity credentials will not magically create or improve a nationwide provider directory. Building that will require stitching disparate APIs and systems together in a standardized way.

e. How could digital identity implementation improve provider data flow?

In theory, a standardized view of identity would allow for simplified authentication, which should lead to improved provider data flow.

f. Would combining FHIR addresses and identity improve data flow?

Yes - if a digital identity credential could be resolved to a unique FHIR address, this would help improve data flow.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

Standardizing authentication mechanisms via trust credentials, as proposed, would better support the goals of reducing patient and provider burden while also improving identity management and security.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

We do not have access to data on the utilization of the information blocking exceptions or conditions with exceptions - if these are being utilized heavily to block information flow, then ASTP / ONC should certainly consider removing these sections. We have not seen these exception cases used to prevent data flow in our interoperability integrations.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

In our view, the most effective disincentive would be an inverted incentive - rather than penalize provider groups for not sharing information, reward them for participating.

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Better publicizing the portal and making it even easier to submit blocking complaints (via EHR integration?) would encourage broader usage and utilization - the data published by the ASTP / ONC shows that the overwhelming majority of information blocking reports are submitted by patients. Broader negative publicity for bad actors would shed light on the situation and plausibly advance data exchange for the better.

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

One of the biggest obstacles to working in the CMS / Medicare / Medicaid space are the number of regulatory hurdles, ambiguous pathways to financial reimbursement, and latent complexities of implementations. To address these points, we suggest that CMS create pathways to efficiently clear regulatory hurdles and certify applications, with clear reimbursement pathways that are stable and can be relied upon. We also suggest that CMS continue to push on standards-based interoperability as a way to reduce complexity and integration overhead.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

The most valuable data that is not presently available through CMS would include clinical data, encounter / visit notes, imaging, and cost data.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

The most valuable data alongside Blue Button 2.0 includes data procured through health information exchanges, medication adherence data (via SureScripts or equivalent), and biometric data from wearables, etc.

c. What obstacles prevent accessing these data sources today?

These data sources are disparate, isolated, and not necessarily standards-compliant, with integration challenges and quirks.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

We believe that access and convenience are key components to improving health outcomes, so emphasizing or enforcing APIs supporting virtual care would be beneficial to patients and providers.

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

Digital identity implementation would convey security and standardization benefits on the part of patients and application developers. On the flip side, there are ergonomic and usability concerns for patient populations. In particular, in reading through the current comment submissions for this RFI, there appears to be significant patient concern over the digital “tracking” of patients and patient identity.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

As noted above, digital identity credentials confer cybersecurity benefits and simplify data exchange by eliminating one-off and bespoke auth solutions in favor of more standard approaches. However, while some patients will be excited to opt in, many will opt out.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

As OpenID Connect is a very popular and standard authentication mechanism, this would have a positive impact on interoperability in general and reduce integration complexity / variety.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

We suggest continued emphasis on embracing FHIR and downstream-of-FHIR APIs.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

This would significantly improve discoverability, which is a key determinant of access. We suggest that the CMS maintain this directory and take on the cost burden.

TD-6. What unique interoperability functions does TEFCA perform?

TEFCA publishes and maintains standards and frameworks that allow for health information exchange and interoperability.

a. What existing alternatives should be considered?

There are existing information exchanges that remain quite popular and see good utilization, like CareQuality, CommonWell, and the information exchange utilities that layer on top of them.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

Yes, on two levels. One is that there are duplicative elements of TEFCA rules, FHIR spec standards, and FHIR profiles like USCDI. Another is that many exchanges continue to operate on older information exchange protocols and frameworks, like HL7, which is redundant and should be modernized.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

US Core imposes some order on the otherwise very flexible FHIR specification, particularly as it pertains to normative resources like patients, observations, medications, etc. However, the spec is not entirely comprehensive, and is of course not enforced.

a. Does it contain the full extent of data elements you need?

Not entirely - primary omissions include clinical diagnostic data and biometric data.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

The format, being derived from FHIR, is not the issue - this is primarily an issue of adoption and utilization.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Additional data elements would definitely add value - the scoping challenges can be addressed via phased roll-outs.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

We believe that there are sufficient fallbacks to unstructured formats already present in USCDI and FHIR, and so additional defined unstructured formats are unnecessary. In general, structured data can always be processed into unstructured data, but the reverse is much more difficult. Ensuring that the standard contains the minimum viable set of necessary data fields for something to be meaningful, with free-text and unstructured fallbacks, is crucial.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

The most effective certification criteria and standards are the requirement to use FHIR APIs and emphasis on interoperability and integration.

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Prioritizing API-based capabilities allows for more types of companies and products to enter the space, which will drive innovation and reduce the reliance on a few crucial vendors.

b. What would be the drawbacks?

On that same token, API-based certifications are more complicated to evaluate and implement, as now you are certifying based on theoretical programmatic capabilities, not product capabilities. This is mitigated by the fact that a company seeking certification will likely have to show some product capabilities in order to certify the API.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

This will have to be a mandate to use modern APIs (like FHIR), apply profiles (like USCDI) to enforce some basic data formats (even when the underlying data is unstructured), and allow for fallbacks to binary data (like images or PDFs) for records that are not digitized.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

CMS could choose to implement incentives (for comprehensive, timely responses to requests) and penalties (for the opposite) for providers.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

Utilizing bulk data transfer (assuming the utilization of standard FHIR APIs and data formats) would allow for CMS to uniformly compute quality performance data from the source, rather than have this reported by provider groups. This would be a fairly significant undertaking on the part of CMS.

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

ASTP/ONC should consider enforcing that the standard of interchange for these APIs be FHIR - this would directly address the access, exchange, and “use without special effort” clauses in a convenient, standards-based, and interoperable manner. Resources to promote this would also be a tactical lever to speed adoption.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or “EHI export” for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

Yes - this should be standardized to a US Core-compliant FHIR bundle.

b. Are there specific workflow aspects that could be improved?

The patient consent step, in particular, is a key area for improvement.

c. Should CMS consider policy changes to support this capability's use?

Yes - in particular, imposing stricter time limits for responses to requests, and / or mandating that this be real-time and patient initiated.

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

Explicitly supporting other information exchanges and networks would be a clarifying point for companies attempting to divine which networks would work best for their use cases. Quality and quantity of data, quality and quantity of APIs, and integration ergonomics would all be criteria to consider.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

This would enable a level of patient understanding and care personalization that has thus far only been possible in concierge care settings. With many more levers to understanding and summarizing vast volumes of information (like large language models), this will directly improve practice efficiencies and quality and thus patient outcomes.

a. What are the primary obstacles to this?

The biggest obstacles to this are integration complexity and insufficient participation in the exchanges that would support this. This would be mitigated by continuing to enforce standards-based APIs (to simplify integration complexity) and continuing to require participation in the exchanges.

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

USCDI has a very clearly defined and well-articulated structure, which breaks down when something falls outside its purview. A true, full EHI realistically has much less structure (thus requiring much more effort to process and parse) but is likely more complete. Given advances in data processing capabilities, there is a ton of value in having access to a full EHI, but structured data will continue to have value for more sophisticated use cases.

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

We are in the process of connecting our systems to a health information exchange vendor that will write back to CareQuality and CommonWell. We cover our scope of service (telemedicine) nationwide. When complete, we will be searchable by NPI and organization ID. We also have deployed a handful of direct connections to health systems.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

For inbound connections, FHIR. For outbound connections, we use a mixture of FHIR, HL7, and proprietary APIs (not preferred but sometimes required).

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

Yes - see above, we attempt to read and write information uniformly (via intermediary vendor) to both CareQuality and CommonWell.

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

Bulk FHIR would allow for efficiently pulling and pushing a patient's full record (Patient/$everything) into and out of shared networks. This will help build a better global understanding of health at an individual and population level.

b. What are the potential disadvantages of their use?

As with most of these questions, the potential disadvantages are privacy and data (and data processing) complexity.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

A point-to-point model allows for very rigidly controlled data flow, but makes any comprehensive integration a complex N-part problem. A shared network has less restrictions on data flow, but makes every member of the network's job easier.

a. Do current rules encourage scalable network participation?

To an extent, yes - there is pretty active participation in most of the biggest-name networks. But even entities that participate in those networks also often fall back to custom APIs, which are not scalable or generalizable.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

The suggested changes (API unification + reciprocal access requirements) would help improve alignment.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

CMS and ASTP / ONC should incentivize participation and reimburse accordingly, and / or implement penalties for non-compliant organizations.

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

The most common practices that we experience include fees to access data, arcane custom implementation requirements that cannot be reused in any way, and incomplete implementation of standards that prevent smooth transfer of information. This is in addition to situations where data requests go entirely ignored.

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

As mentioned above, incentives (for organizations that share information well) and penalties (for organizations that engage in information blocking) would be useful to encode in policy.

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

We do not believe so.

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

Realistically, patients still do not really get price transparency on things outside of estimated prescription costs. This can be ascribed to complicated data formats and very slow updates.

b. Which workflows would benefit most from functional price transparency?

Price estimation for procedures, referrals, and preventative care would in our opinion be the most valuable areas to focus workflow improvements.

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

The ultimate general-purpose solution that benefits all parties equally would be a comprehensive price transparency API (across all relevant domains, including Rx, visits, procedures, etc).

d. What would further motivate solution development?

Adding reimbursement incentives to groups that participate in and provide price transparency would help motivate solution development in this area.